AETRAA · BILL OF QUANTITIES TOOLS

How to use the BOQ tools, one click at a time

This copy is for a **Mac**. It shows every click. You do not need to know anything about computers or code.

TWO TOOLS, ONE FOLDER

Tool 1, the BOQ agent: you give it a folder of drawings. It gives back the finished bill (BOQ), the take-off, coloured drawings and a list of questions.

Tool 2, the platform: a page in your web browser. You upload drawings, look at the results, correct them and measure by hand.

Before you start, have these ready

- **The zip file** `AETRAA for Ajay.zip`, sent to you by Girish.
- **Your drawings** for one project, as PDF files, in one folder. Folders inside folders are fine. One big PDF with many sheets is fine too.
- **Internet**, the first time only. Each tool sets itself up on its first run.
- **Excel** or Numbers to open the results.

PART 1

Put the files on your computer

You do this once. When a new version arrives, see “Getting a new version” at the end of this part.

1. Click the **Finder** icon in the Dock at the bottom of the screen. It is the blue and white smiling face.

2. In the list on the left of the Finder window, click **Downloads**.

   TIP
   If you saved the zip somewhere else, such as from Google Drive or WhatsApp, open that place instead.

3. Find `AETRAA for Ajay.zip` and **double-click** it.

   YOU WILL SEE
   A new folder called `AETRAA` appears next to the zip. This takes a few seconds.

4. **Drag** the `AETRAA` folder onto **Documents** in the list on the left, then let go.

   YOU WILL SEE
   The folder now lives in Documents. From now on, open it from there.

What is inside the AETRAA folder

You only need the files marked in blue. Leave everything else alone.

```
AETRAA
└── Girish
    ├── sdk agent - final file        ← Tool 1, the BOQ agent
    │     ├── Run agent (Mac).command     ← double-click this to make a bill
    │     ├── output                  ← your results appear here
    │     └── readings                ← the agent's memory. Never delete this
    └── boq_arch                      ← Tool 2, the platform
          ├── Start platform (Mac).command      ← double-click this to open th
```

Getting a new version later

1. Open your old `AETRAA → Girish → sdk agent - final file` folder. Copy the `readings` folder to your Desktop, so it is safe.

2. Delete the old `AETRAA` folder. Then put the new one in its place by following steps 1 to 4 above.

3. Drag the `readings` folder from your Desktop back into the new `sdk agent - final file` folder. If you are asked, choose **Replace**.

IMPORTANT

If you used an older copy in `Documents/AETRAA/agents/boq_pomi_v1`, stop using it. Copy its `readings` folder into the new `sdk agent - final file` folder and use only the new one. The old copy has none of the latest fixes.

PART 2

The “are you sure?” message

The first time you double-click one of the tool files, your computer checks whether you trust it. This is normal. You do it once for each file.

1. Double-click the file, for example `Run agent (Mac).command`.

   YOU WILL SEE
   A message saying the file can’t be opened because Apple could not check it. Click **Done** or **OK**.

2. Click the **Apple logo** at the very top-left of the screen, then click **System Settings…**

3. In the list on the left, click **Privacy & Security**. Scroll down on the right until you see a line saying the file “was blocked”.

4. Click **Open Anyway**. Type your Mac password if asked, then click **Open**.

   YOU WILL SEE
   A Terminal window opens and the tool starts. From now on, a double-click is enough.

OLDER MACS
If you don’t see “Open Anyway”, try this instead. Hold down the **control** key, click the file, and choose **Open**. Then click **Open** again in the message.

If a text file opens instead, or it says “permission denied”

1. Press **command** + **space**. Type **Terminal** and press **return**.

2. In the Terminal window, type `chmod +x ` . That is “chmod”, a space, “+x”, and another space.

3. Drag the tool file from Finder into the Terminal window. Its name appears after what you typed. Press **return** .

4. Close Terminal and double-click the file again.

If it says “Python 3.10 or newer is needed”

1. Open **Safari** and go to **python.org/downloads**.

2. Click the big yellow **Download Python 3.x** button. Any version from 3.10 up works.

3. Open **Downloads** in Finder and double-click the file you just downloaded. Its name starts with `python-` .

4. Click **Continue** , **Continue** , **Agree** , then **Install** . Type your Mac password and wait. Click **Close** when it is done.

5. Double-click the tool file again.

PART 3 · TOOL 1

Make a bill with the BOQ agent

You point it at a folder of drawings and it writes the bill. It reads the drawings, counts and measures them, and fills in the POMI bill.

1. Put all the drawings for **one** project in **one** folder, for example on your Desktop. They must be PDF files.

2. Open `AETRAA` → `Girish` → `sdk agent - final file`.

3. Double-click `Run agent (Mac).command`.

   YOU WILL SEE
   A window full of text opens: Terminal. The first time, it says **“Installing what the agent needs”**. Wait. This takes a few minutes, once.

4. A box pops up: **“Choose the folder that holds the drawings”**. Click your drawings folder **once**, then click **Choose**.

   TIP
   Choose the folder itself, not a file inside it.

5. **Wait, and don’t close the window.** You will see lines appear one by one, from `S0 Ingest` to `S21 Revise`. Each line is one step of the work.

   HOW LONG?
   A few sheets take a few minutes. A big set of 300 sheets or more can take 20 to 40 minutes or longer. Keep the computer plugged in and awake.

6 When it finishes, it prints **“Open this folder: …”** and the `output` folder opens by itself.

In Terminal, press **return** to finish. You can then close the window.

7 In the `output` folder, open the folder named after your project, for example `RIVERSIDE CLINIC - Rev 00`.

What you get, and what to open first

FILE	WHAT IT IS
Questions and gaps.txt	**Open this first.** It lists everything the drawings did not answer, one line each, such as a count that differs from the plans.
… BOQ … .xlsx	The bill of quantities, with and without prices.
… Take-off … .xlsx	Every measured line, with its sizes and the drawing it came from.
markup (folder)	The drawings with every measured item coloured in, and a legend that explains the colours.
Provenance … .xlsx	Where every number came from.
Drawing register.xlsx	Every sheet in the set, with its number and title.
WHY THERE IS NO BILL.txt	This appears only when the bill was held back. It says why in plain words. Go to Part 4.

RUNNING IT AGAIN

If you run the same drawings again, you get the same revision. If you change the drawings and run again, the bill becomes Rev 01, then Rev 02, and so on.

PART 4

When the bill is held back

Sometimes a door or window schedule is drawn as a picture instead of text. The agent does not guess what a picture says. It either reads it with AI, or it waits for someone to read it.

1. Open **WHY THERE IS NO BILL.txt** in the output folder. It names the sheet that could not be read, for example **“A-501 not read”**.

2. The agent has left a note for that sheet in the `readings` folder. Its name ends in `.request`.

3. **The easy way:** open the `sdk agent - final file` folder in **VS Code**. Open the **Claude** panel and type:

   ```
   answer the requests, then run the BOQ again on "<your drawings folder>"
   ```

   Claude reads each drawn schedule, copying exactly what is printed and measuring nothing. Then it runs the agent again and tells you the result.

   GETTING THE FOLDER’S LOCATION

   In Finder, hold **option** and right-click the folder, then choose **Copy … as Pathname**. Paste it between the quote marks.

4. **No VS Code?** Send the **WHY THERE IS NO BILL.txt** file to Girish.

WHY THIS HAPPENS LESS WITH AI SWITCHED ON

If Claude is installed and signed in on this computer, the agent reads drawn schedules by itself during the run, and you never see a request. Without it, the agent still works, but it has to leave these notes.

PART 5 · TOOL 2

Use the platform in your browser

The platform is a page in your web browser. You upload drawings, run a take-off, look at the results, and correct or measure things by hand.

Open it

1. Open `AETRAA` → `Girish` → `boq_arch`.

2. Double-click `Start platform (Mac).command`.

   YOU WILL SEE
   A window full of text opens. The first time, it installs the platform's parts. **This can take 5 to 10 minutes.** Wait until it says **"Starting …"**.

3. Your web browser opens on the sign-in page by itself.

   IF THE PAGE SAYS IT CAN'T BE REACHED
   The platform is still starting. Wait 10 seconds and click the browser's refresh button ⟳.

4. Type the username and password exactly as shown, all in small letters, then click **Sign in**.

   Username `ajay123` Password `aetraa123`

KEEP THAT WINDOW OPEN
The Terminal window is the platform itself. If you close it, the platform stops. Only close it when you have finished for the day.

Make a take-off

1. Under **“Add the drawings”**, click **Click to choose**.

2. Select your drawing files. To select many at once, hold **command** and click each one. You can also choose a single `.zip` of the drawings. Then click **Open**.

   TIP
   You can also drag the files from Finder straight onto that box. Upload the **whole** architectural set together: plans, sections, schedules and details.

3. Leave the other boxes as they are. Scroll down and click **Run take-off**.

   YOU WILL SEE
   A progress page. The work is written out line by line as it happens. Wait until it finishes.

4. When it finishes, you reach the results page. Under **“Your files”**, each result has three buttons:

   - **Open** shows it right in the browser.
   - **Download** saves the Excel file to your computer.
   - **With drawings (.zip)** saves the Excel file together with its drawings, so the drawing links still work when you send it to someone.

Check, correct and measure by hand

1. On the results page, click the **Measure** tab.

2. The first time, a short guided tour opens. You can open it again at any time with **? How to measure** at the top right.

3 Use the tools along the top. The most useful ones:

TOOL	KEY	HOW TO USE IT
Length	L	**Double-click** to start, click at each corner, **double-click** to finish.
Area	A	The same as Length, but going around the shape.
Count	C	Click on each item. Click a ring again to remove it.
Count similar	S	Draw a box around one symbol and it finds the others like it. Then check the rings it placed.
Set scale	K	Click two points a known distance apart and type that distance.
Eraser	E	Point at something (it turns red) and click to remove it.
Undo	⌘+Z	Takes back the last thing you did.

MOVING AROUND

Scroll to zoom in and out. Drag with the mouse to move the drawing. Press **F** to see the whole sheet.

4 When you have finished, click **Export to Excel**. Your hand measurements are added to the take-off and the bill. The automatic results are kept as they were.

Close it

Click the Terminal window, press **command** + **Q**, and click **Terminate** if asked. Your runs are saved.

PART 6

Share the platform with someone

This lets someone use the platform from anywhere through a web link, while it runs on one Windows PC.

THIS IS DONE ON THE WINDOWS PC

Sharing is set up on Girish's Windows PC. If Girish sends you a link, you only need to follow the "If someone sent you a link" steps below.

If someone sent you a link

1. Click the link. It opens the sign-in page in your browser.
2. Type **ajay123** and **aetraa123**, then click **Sign in**.
3. Carry on from "Make a take-off" in Part 5. Everything works the same.

PART 7

If something goes wrong

WHAT YOU SEE	WHAT TO DO
“Windows protected your PC”, or “Apple could not verify”	Follow Part 2. You do this once for each file.
Double-clicking opens a text file	Do the “permission denied” steps in Part 2.
“No folder chosen”	You clicked Cancel. Double-click the file again and choose the folder.
“Install failed” or “did not install”	Check the internet connection, then double-click the file again.
It has been running for a long time	Big sets take 20 to 40 minutes or more. Leave the window open. It is still working if new lines keep appearing.
“No deliverable issued”, or **WHY THERE IS NO BILL.txt**	See Part 4.
In the gaps: “password-protected”	Ask the consultant for that drawing without a password.
In the gaps: “damaged or is not a PDF”	Ask the consultant to send that drawing again.
In the gaps: “Image only … Measurement by hand”	That sheet is a scan. Measure it by hand in the platform’s Measure tab (Part 5).
The bill says “DO NOT ISSUE – not recalculated”	The numbers are in the bill, but the totals were not recalculated. Install the free program **LibreOffice** from libreoffice.org, then run again.
The browser says the page can’t be reached	Wait 10 seconds and refresh. Check that the platform’s window is still open.

“Wrong username or password”	Type **ajay123** and **aetraa123** in small letters, with no spaces.
Anything else	Take a photo or screenshot of the window and send it to Girish. To take a screenshot, press **shift** + **command** + **4**, then drag a box around the window.

Words you will see

BOQ	Bill of Quantities: the list of every item to build, with how many of each.
Take-off	The working sheet behind the bill, showing how each quantity was measured.
Markup	A drawing with the measured items coloured in.
Gap	A question the drawings did not answer. It is listed rather than guessed.
Rev 00, Rev 01	The revision: the first issue of the bill, then the next one, and so on.
POMI	The standard way of measuring that the bill follows.
